

MPC-PiNNs technique applied to Quadrotors

Elective in Robotics [UAV] Course Project

Master's Degree in Artificial Intelligence and Robotics

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Unicycle

2 Models

$$\begin{cases} \dot{x} = v \cos(\theta) \\ \dot{y} = v \sin(\theta) \\ \dot{\theta} = w \end{cases} \quad (1)$$



Quadorotor

2 Models



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PiNNs

4 Physical Informed Neural Networks

Physical Informed Neural Networks (PiNNs) propose an alternative approach where the solution is approximated by a neural network that is trained not only on data but also by enforcing the underlying physical laws via the differential equations.

$$\mathcal{L}_{\text{data}}(\theta) = \frac{1}{N_u} \sum_{i=1}^{N_u} |u_{\theta}(\mathbf{x}_i, t_i) - u_i^{\text{obs}}|^2 \quad (2)$$

$$\mathcal{L}_{\text{phys}}(\theta) = \frac{1}{N_r} \sum_{j=1}^{N_r} |\mathcal{N}[u_{\theta}](\mathbf{x}_j, t_j)|^2, \quad (3)$$



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